

ETC3530 - Assignment Report

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Inputs: from question 2 to question 4

- **Issue age:** $x=40$ (male).
- **Term:** 20 years; ceases at death or end of policy year 20.
- **Benefit timing:** Payable **end of year of death**.
- **Sum assured at year k :** $S(1 + c)^k$
 - with $S=150,000$ **annual increase** $c=0.025$
- **Premiums:** Level, **payable in advance** while in force, max 20 payments.
- **Expenses:**
 - **Initial:** $E_0 = 250$ at time 0
 - **Renewal:** $e=400$ at start of each year while in force.
- **Per-claim expense:** **None** unless explicitly introduced in Q4(v) (then $c_{\text{claim}} = 250$ at claim time).
- **Interest rate:** $i=0.04$
- **Discount factor:** $v = (1 + i)^{-1} = 0.961538462$
- **Mortality:** ALT 2020–22
- **Selection/Improvements:** No mortality improvement; use table as given.
- **Inflation:** Not modeled (except the contractual benefit escalation via c).
- **Taxes/Profit-sharing:** Not modeled.

Question 1 — Survival and death measure

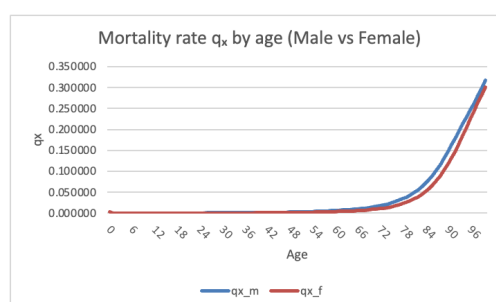
$$\text{i) } q_x = 1 - \frac{l_{x+1}}{l_x} = \frac{l_x - l_{x+1}}{l_x}$$

$$\text{ii) } {}_k p_x = \frac{l_{x+k}}{l_x}$$

$$\text{iii) } {}_k | q_x = \frac{l_{x+k} - l_{x+k+1}}{l_x}$$

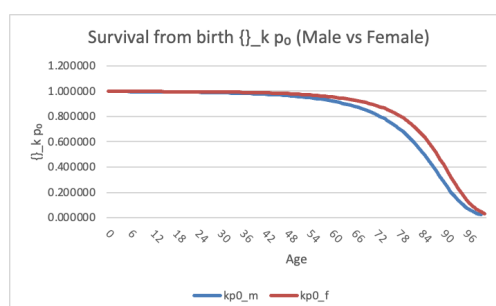
Mortality rate q_x :

Mortality is close to 0 at young ages, rises slowly through mid-life, then increases steeply from about age 70 onward and approaches 1 at extreme ages. At every age females have lower q_x than males, with the gap widening in older ages. This explains why female survival curves sit above male curves.



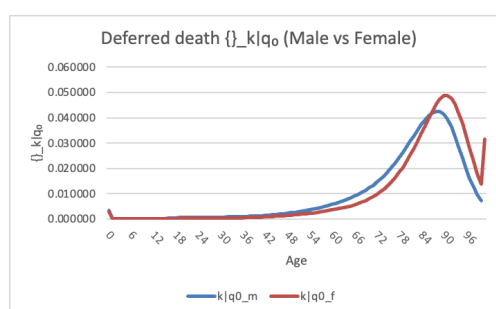
Survival ${}_k p_0$:

Survival decreases monotonically with age (starts at 1 at birth towards 0 at extreme ages). The female curve is consistently above the male curve, showing higher survival probabilities at all ages, with differences most visible after age 60. This translates into higher longevity risk for females—longer annuity payout periods and a higher chance of outliving retirement savings.



Deferred death ${}_k | q_0$:

Very low at young ages, then rises rapidly, peaking around ages 88–92 for females—higher and later than the male peak (~86–90)—before declining as few lives remain beyond 95–100. Although females have lower (q_x) at each age, more females survive to very old ages, so the distribution of “death in year (k)” shifts rightward and can peak higher for females. For insurers, this concentrates expected claim frequency in the late-old ages, producing a claims “bulge” for final-expense and whole-of-life products.



Life expectancy:

At birth ($E[K_0]$), life expectancy is 80.77 years for males vs 84.75 years for females, a gap of about four years. At age 60 ($E[K_{60}]$), it is 23.92 vs 26.76 years, with females still ahead by nearly three years.

Conclusion: Females experience lower age-specific mortality and higher survival across all ages, leading to consistently longer life expectancies. Mortality accelerates steeply at older ages, which is visible in the sharp rise in q_x and the peak of ${}_k|q_0$ in the late 80s–90s.

For business practice, this underpins gender differences in pricing, reserving, and pension planning, and highlights the importance of incorporating realistic mortality patterns into insurance and retirement product design.

Question 2 - Pricing an increasing 20-year term assurance

ii) Compute $\mathbb{E}[B_x(0)]$

$$\begin{aligned}\mathbb{E}[B_x(0)] &= \sum_{k=0}^{19} v^{k+1} S(1+c)^k {}_k|q_{40} = S \sum_{k=0}^{19} (1+i)^{-k-1} (1+c)^k \frac{l_{40+k} - l_{40+k+1}}{l_{40}} \\ &= 150000 \sum_{k=0}^{19} 0.9615^{k+1} (1+0.025)^k \frac{l_{40+k} - l_{40+k+1}}{l_{40}} = 6,989.14\end{aligned}$$

iv) Give formulas for $\mathbb{E}[E_x(0)]$ and $\mathbb{E}[G_x(0)]$

$$\begin{aligned}\mathbb{E}[E_x(0)] &= E_0 + e(\ddot{a}_{40:\overline{20}} - 1) = 250 + 400 \sum_{k=1}^{19} v^k {}_k p_{40} \\ &= 250 + 400 \sum_{k=1}^{19} 0.9615^k {}_k p_{40} = 5,408.83 \\ \mathbb{E}[G_x(0)] &= g_{\text{gross}} \ddot{a}_{40:\overline{20}} = g_{\text{gross}} \sum_{k=0}^{19} v^k {}_k p_{40} = g_{\text{gross}} \sum_{k=0}^{19} 0.9615^k {}_k p_{40}\end{aligned}$$

v) Compute $\mathbb{E}[F_x(0)]$

$$\begin{aligned}\mathbb{E}[F_x(0)] &= \mathbb{E}[B_x(0)] + \mathbb{E}[E_x(0)] - \mathbb{E}[G_x(0)] = \mathbb{E}[B_x(0)] + \mathbb{E}[E_x(0)] - g \cdot \ddot{a}_{40:\overline{20}} \\ &= 6,989.14 + 5,408.83 - 13.897 g_{\text{gross}} = 12,397.97 - 13.897 g_{\text{gross}}\end{aligned}$$

$$\begin{cases} \mathbb{E}[F_x(0)] > 0 & \implies \text{expected loss} \\ \mathbb{E}[F_x(0)] = 0 & \implies \text{break-even, } g_{\text{gross}} = g_0, \text{ (principle of equivalence)} \\ \mathbb{E}[F_x(0)] < 0 & \implies \text{expected gain} \end{cases}$$

→ At the chosen gross, if $g_{\text{gross}} = g_0$ (break even premium), the product is expected to break even (neither profit nor loss) - the expected present value of premiums exactly matches the expected present value of benefits and expenses. If $g_{\text{gross}} > g_0$, the product is expected to make a profit. If $g_{\text{gross}} < g_0$ expected it is expected to make a loss.

vi) Derive and implement the break-even premium g_0

$$g_0 = \frac{\mathbb{E}[B_x(0)] + \mathbb{E}[E_x(0)]}{\ddot{a}_{40:\overline{20}}} = \frac{(6,989.14 + 5,408.83)}{13.897} = 892.13$$

vii) Target profit: for $\pi = \$7,000$ (EPV at time 0), derive and compute g_π

$$\text{Profit} = \pi \Rightarrow \mathbb{E}[F_x(0)] < 0 \text{ and } \mathbb{E}[F_x(0)] = \pi \rightarrow \mathbb{E}[F_x(0)] = -\pi$$

$$g_\pi = \frac{\mathbb{E}[B_x(0)] + E_0 + \pi}{\ddot{a}_{40:\overline{20}}} + e = \frac{6,989.14 + 250 + 7000}{13.897} + 400 = 1,395.83$$

viii) How i and c affect $\mathbb{E}[B_x(0)]$ and g_0

Param	Value	EPV_B
i	0.03	7,902.46
i	0.05	6,206.16
c	0.015	6,249.30
c	0.035	7,830.83

Higher i lowers EPV(B) because future death benefits are discounted more.

Since A ($\ddot{a}_{40:\overline{20}} = 13.897$) also falls when i rises, the reduction in EPV(B) dominated. As a result, the break-even premium on g_0 is still downward when interest rates increase. Intuitively, in higher-rate environments the insurer can invest premiums at better returns, while future death benefits lose more of their present value, so a lower premium is sufficient to balance the contract.

By contrast, increasing the annual increase c raises EPV(B) because later-year benefits are larger. The annuity factor A is almost unchanged, so the higher benefit PV pushes up the numerator in the premium equation. Consequently, g_0 rises with c. In practical terms, a higher escalation rate backloads the protection, concentrating more risk in later durations. To cover this heavier expected liability, the insurer must charge a higher premium.

Question 3 - Prospective reserves and recursion

i) Write $\mathbb{E}[B_x(k) \mid K_x \geq k]$ for the term contract

$$\mathbb{E}[B_x(k) \mid K_x \geq k] = \sum_{j=0}^{19-k} v^{j+1} S(1+c)^{k+j} {}_j|q_{x+k}$$

$$\text{Whole-of-life case : } \mathbb{E}[B_x(k) \mid K_x \geq k]_{\text{whole-of-life}} = \sum_{j=0}^{\infty} v^{j+1} S(1+c)^{k+j} {}_j|q_{x+k}$$

Whole-of-life has no term cap, so the summation continues indefinitely.

ii) Prove the identity ${}_j|q_{x+k} = q_{x+k}$ for $j = 0$ and ${}_{(j-1)}|q_{x+k+1}p_{x+k}$ for $j \geq 1$

$${}_j|q_{x+k} = \frac{l_{x+k+j} - l_{x+k+j+1}}{l_{x+k}}$$

$$\text{with } j=0: {}_0|q_{x+k} = \frac{l_{x+k} - l_{x+k+1}}{l_{x+k}} = q_{x+k}$$

$$\text{with } j \geq 1 : {}_j|q_{x+k} = \frac{l_{x+k+j} - l_{x+k+j+1}}{l_{x+k}} = \frac{l_{x+k+1+j-1} - l_{x+k+j-1+1+1}}{l_{x+k+1}} \cdot \frac{l_{x+k+1}}{l_{x+k}} \\ = (j-1)|q_{x+k+1} \cdot p_{x+k}$$

iii) Recursion for benefits:

$$\mathbb{E}[B_x(k+1) \mid K_x \geq k+1] = \sum_{j=0}^{19-(k+1)} v^{j+1} S(1+c)^{k+1+j} {}_j|q_{x+k+1} \quad (1)$$

From ii, we have: ${}_{t-1}q_{x+k+1} \cdot p_{x+k} = {}_t|q_{x+k}$

$$\text{Let } j = t-1 \rightarrow t = j+1 \rightarrow {}_j|q_{x+k+1} = \frac{{}_{j+1}|q_{x+k}}{p_{x+k}}$$

$$\text{From (1): } \mathbb{E}[B_x(k+1) \mid K_x \geq k+1] = \frac{1}{p_{x+k}} \sum_{j=0}^{18-k} v^{j+1} S(1+c)^{k+j+1} {}_{j+1}|q_{x+k}$$

$$\mathbb{E}[B_x(k) \mid K_x \geq k](1+i) = (1+i) \sum_{j=0}^{19-k} v^{j+1} S(1+c)^{k+j} {}_j|q_{x+k}$$

$$= \sum_{j=0}^{19-k} v^j S(1+c)^{k+j} {}_j|q_{x+k} \quad (2)$$

$$\text{At } j=0: (2) = v^0 S(1+c)^k {}_0|q_{x+k} = S(1+c)^k q_{x+k}$$

$$\rightarrow (2) = S(1+c)^k q_{x+k} + \sum_{j=1}^{19-k} v^j S(1+c)^{k+j} {}_j|q_{x+k} \quad (3)$$

$$\text{As } \sum_{j=1}^{19-k} (j) = \sum_{j=0}^{18-k} (j+1)$$

$$(3) = S(1+c)^k q_{x+k} + \sum_{j=0}^{18-k} v^{j+1} S(1+c)^{k+j+1} {}_{j+1}|q_{x+k}$$

$$\rightarrow \sum_{j=0}^{18-k} v^{j+1} S(1+c)^{k+j+1} {}_{j+1}|q_{x+k} = \mathbb{E}[B_x(k) \mid K_x \geq k] \cdot (1+i) - S(1+c)^k q_{x+k}$$

$$\text{So } \mathbb{E}[B_x(k+1) \mid K_x \geq k+1] = \frac{1}{p_{x+k}} \cdot (\mathbb{E}[B_x(k) \mid K_x \geq k] \cdot (1+i) - S(1+c)^k q_{x+k}) \\ \text{with } \mathbb{E}[B_x(20) \mid K_x \geq 20] = 0$$

iv) Recursion for premiums:

$$\mathbb{E}[G_x(k+1) \mid K_x \geq k+1] = \sum_{j=0}^{18-k} v^j g {}_j p_{x+k+1}$$

$$\text{As: } {}_{j+1}p_{x+k} = \frac{l_{x+k+j+1}}{l_{x+k}} = \frac{l_{x+k+j+1}}{l_{x+k+1}} \cdot \frac{l_{x+k+1}}{l_{x+k}} = {}_j p_{x+k+1} \cdot p_{x+k}$$

$$\text{So } \mathbb{E}[G_x(k+1) \mid K_x \geq k+1] = \frac{g}{p_{x+k}} \sum_{j=0}^{18-k} v^j {}_{j+1}p_{x+k} \quad (4)$$

$$\mathbb{E}[G_x(k) \mid K_x \geq k] = \sum_{j=0}^{19-k} v^j g {}_j p_{x+k}$$

$$\text{At } j=0 \rightarrow \mathbb{E}[G_x(k) \mid K_x \geq k] = g$$

$$\begin{aligned}
&\rightarrow \mathbb{E}[G_x(k) \mid K_x \geq k] = g + g \sum_{j=1}^{19-k} v^j p_{x+k} \\
&\rightarrow \sum_{j=1}^{19-k} v^j p_{x+k} \cdot (1-i) = \frac{\mathbb{E}[G_x(k) \mid K_x \geq k] - g}{g} \cdot (1-i) = \sum_{j=0}^{18-k} v^{j+1} p_{x+k}. \quad (5)
\end{aligned}$$

From (4) & (5),

$$\begin{aligned}
\mathbb{E}[G_x(k+1) \mid K_x \geq k+1] &= \frac{1}{p_{x+k}} \cdot (\mathbb{E}[G_x(k) \mid K_x \geq k] - g) \cdot (1-i), \\
&\text{with } \mathbb{E}[G_x(20) \mid K_x \geq 20] = 0
\end{aligned}$$

v) Recursion for expenses

$$\begin{aligned}
\mathbb{E}[E_x(k+1) \mid K_x \geq k+1] &= \frac{1}{p_{x+k}} \cdot (\mathbb{E}[E_x(k) \mid K_x \geq k] - e) \cdot (1-i) \\
e_k &= \begin{cases} E_0, & \text{if } k = 0, \\ e, & \text{if } 1 \leq k \leq 19, \\ 0, & \text{if } k = 20. \end{cases} \quad \text{with } \mathbb{E}[E_x(20) \mid K_x \geq 20] = 0
\end{aligned}$$

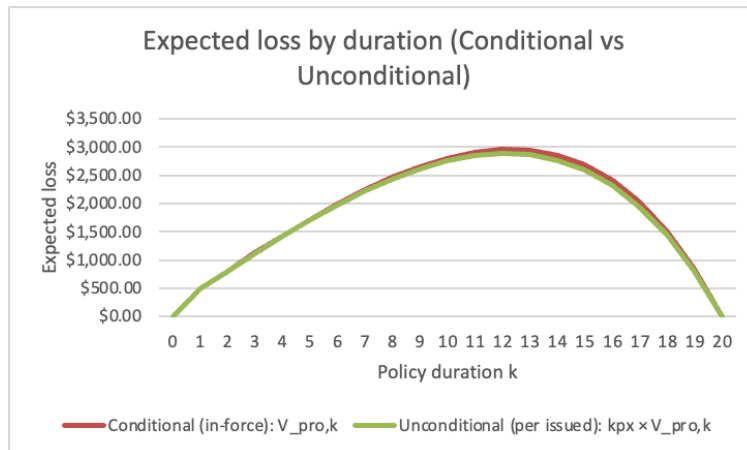
vi) Reserve Recursion:

$$\begin{aligned}
V_{k+1}^{\text{pro}} &= \mathbb{E}[B_x(k+1) \mid K_x \geq k+1] - \mathbb{E}[G_x(k+1) \mid K_x \geq k+1] + \mathbb{E}[E_x(k+1) \mid K_x \geq k+1] \\
&= \frac{1}{p_{x+k}} \left(\mathbb{E}[B_x(k) \mid K_x \geq k] \cdot (1+i) - S(1+c)^k q_{x+k} - \frac{(\mathbb{E}[G_x(k) \mid K_x \geq k] - g)(1+i)}{p_{x+k}} + \right. \\
&\quad \left. \frac{(\mathbb{E}[E_x(k) \mid K_x \geq k] - e)(1+i)}{p_{x+k}} \right) \\
&= \frac{(1+i)}{p_{x+k}} \cdot (\mathbb{E}[B_x(k) \mid K_x \geq k] - \mathbb{E}[G_x(k) \mid K_x \geq k] + \mathbb{E}[E_x(k) \mid K_x \geq k]) - \frac{S(1+c)^k q_{x+k}}{p_{x+k}}
\end{aligned}$$

$$\text{So } V_{k+1}^{\text{pro}} = \frac{(V_k^{\text{pro}} + g - e)(1+i) - S(1+c)^k q_{x+k}}{p_{x+k}}, \quad V_{20}^{\text{pro}} = 0$$

viii) Different trends of the conditional and unconditional series:

$$\begin{aligned}
\mathbb{E}[F_x(k)] &= {}_k p_x \mathbb{E}[F_x(k) \mid K_x \geq k] \\
\text{As } \mathbb{E}[F_x(k) \mid K_x \geq k] &= V_k^{\text{pro}} \rightarrow \mathbb{E}[F_x(k)] = {}_k p_x V_k^{\text{pro}}
\end{aligned}$$



The conditional series shows the expected future loss per policy in force. It rises in the early years because many years remain for death benefits to occur and the sum assured is increasing, peaks around the middle when mortality and benefit levels are highest, and then declines as the remaining horizon shortens and discounting dominates.

The unconditional series multiplies this reserve by the probability of survival to each duration. In the early years, when survival is almost certain, it closely matches the conditional series. In later years, as fewer policies survive, it falls more quickly and always sits below the conditional line, converging to zero at expiry.

In practice, the conditional measure is most relevant for valuing and managing reserves on the in-force portfolio, while the unconditional measure reflects the average expected loss across all policies issued at outset. For the insurer, this distinction highlights both the reserve strain on surviving contracts and the overall liability profile of the entire book.

Question 4 - Profit emergence and profitability

ii) $(PRO)_k = (CF)_k - (TR)_k = (g_{\text{actual}} - g_{\text{gross}}) \cdot (1 + i)$

$(PRO)_k$ is constant because it only depends on the premium margin $(g_{\text{actual}} - g_{\text{gross}})$ and interest; mortality, survival probabilities, and benefit timing cancel out in the recursion, leaving a flat value each year. In practice, this means the insurer earns the same expected profit each year, since the extra loading in premiums emerges evenly across the contract term.

iii) (P_0) is the EPV of the profit stream; it equals the premium margin $(g_{\text{actual}} - g_{\text{gross}})$ times the annuity value $(\ddot{a}_{40:\overline{20}})$ (since $(PRO)_k$ is flat). Setting the margin to $7000/A$ makes $P_0 \approx \pi = 7000$, so the pricing meets the target profit at time 0. Any small difference is rounding or numerical precision, not a model mismatch. Practically, this means the chosen level premium embeds exactly the desired profit loading, independent of claim timing.

iv) Profitability ratio $f = P_0/I_0$

gactual	P0	I0	f = P0/I0
1395.832	7000	18651.89	0.375297
1500	8447.633	20043.84	0.421458
1750	11921.9	23384.48	0.509821
2000	15396.16	26725.12	0.576093
2250	18870.43	30065.76	0.627639
2500	22344.7	33406.4	0.668875
2750	25818.96	36747.04	0.702613
3000	29293.23	40087.68	0.730729
3250	32767.49	43428.32	0.754519
3500	36241.76	46768.96	0.77491
3750	39716.02	50109.6	0.792583
4000	43190.29	53450.24	0.808047

The EPV of premiums I_0 is linear in the actual premium g_{actual} , since the annuity factor is fixed. The EPV of profit P_0 also rises roughly linearly with g_{actual} , because $(PRO)_k = g_{\text{actual}} - g_0)(1 + i)$ does not depend on duration k . As a result, the profitability ratio $f = P_0/I_0$ increases with higher g_{actual} , moving from about 0.42 at $g=1500$ to about 0.81 at $g=4000$. In practice, this means that if the insurer charges a higher premium, not only does total premium income increase, but the profit margin (as a share of income) also improves. This reflects how premium loading directly strengthens profitability, provided lapse and claim experience follow assumptions.

v) A per-claim expense $c_{\text{claim}} = 250$ is introduced

We treat the \$250 per-claim cost as part of the death benefit

$$\begin{aligned}
\text{EPV}(B)_{\text{new}} &= \sum_{k=0}^{19} v^{k+1} {}_k|q_x (S(1+c)^k + c_{\text{claim}}) \\
&= \text{EPV}(B)_{\text{old}} + c_{\text{claim}} \sum_{k=0}^{19} v^{k+1} {}_k|q_x = \text{EPV}(B)_{\text{old}} + 250A_{40:\overline{20}}
\end{aligned}$$

$$\text{EPV}(E)_{\text{new}} = \text{EPV}(E)_{\text{old}} \quad (\text{not change})$$

$$\begin{aligned}
V_k^{\text{pro}} &= \frac{p_{x+k} V_{k+1}^{\text{pro}} + q_{x+k} (S(1+c)^k + 250)}{1+i} + e - g \\
\Rightarrow V_k^{\text{pro,new}} &= V_k^{\text{pro,old}} + \frac{250 q_{x+k}}{1+i}, \quad V_{20}^{\text{pro}} = 0
\end{aligned}$$

$$g_0^{\text{new}} = \frac{E[B_x(0)] + E[E_x(0)] + 250 A_{40:\overline{20}}}{\ddot{a}_{40:\overline{20}}} = g_0^{\text{old}} + \frac{250 A_{40:\overline{20}}}{\ddot{a}_{40:\overline{20}}}$$

When a per-claim expense $c_{\text{claim}} = 250$ is introduced, the EPV of benefits rises by $c_{\text{claim}} \sum v^{k+1} {}_k|q_x$, since each death now triggers an additional cost. In the reserve recursion, the benefit term becomes $[S(1+c)^k + c_{\text{claim}}]q_{x+k}$. The denominator (the annuity factor for premiums) is unchanged, the new break-even premium g_0 must increase to cover this added liability. From a business perspective, this adjustment mirrors the claims handling and settlement costs that insurers incur in real life (doctor's reports, administration, payments). Even if such expenses seem small compared to the sum assured, they add up across the portfolio. Ignoring them would understate liabilities and erode profitability, so insurers must load them into premiums to keep margins aligned with targets.